CHAPTER 10 GASES

Key idea: gas pressure comes from molecules hitting the walls; temperature measures their mean kinetic energy.

1 IDEAL GAS EQUATION

$$pV = nRT = NkT$$

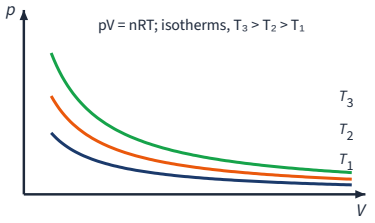
- R 8.31 J K^{-1} mol^{-1}
- k $R/N_A = 1.38 \times 10^{-23}$ J K^{-1}
- N_A 6.02×10^{23} mol^{-1}
- n $= \dfrac{m}{M} = \dfrac{N}{N_A}$

T in kelvin: $T = \theta + 273$

2 GAS LAWS

Law	Constant	Relation
Boyle's	T	pV = constant
Charles's	p	$V \propto T$
Pressure	V	$p \propto T$

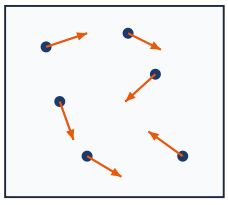
$$\frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}$$



3 KINETIC THEORY ASSUMPTIONS

1. Very many identical molecules in **random** motion
2. Volume of molecules negligible
3. No forces except during collisions
4. Collisions are **elastic**; collision time negligible
5. Newtonian mechanics applies

4 PRESSURE OF A GAS



random motion, elastic collisions

p = ⅓ρ⟨c²⟩

$$p = \tfrac{1}{3}\rho\langle c^2\rangle = \tfrac{1}{3}\frac{Nm}{V}\langle c^2\rangle$$

Pressure = rate of change of momentum of molecules per unit area of wall.

5 MOLECULAR SPEEDS

$$c_{\text{rms}} = \sqrt{\langle c^2\rangle} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}}$$

- $c_{\text{rms}} \propto \sqrt{T}$ and $\propto 1/\sqrt{M}$
- Mean square: square first, then average

6 ENERGY & EQUIPARTITION

$$\langle E_k\rangle = \tfrac{1}{2}m\langle c^2\rangle = \tfrac{3}{2}kT$$

Each degree of freedom: $\frac{1}{2}kT$ per molecule.

Gas	f	U for n mol
Monatomic	3	$\frac{3}{2}nRT$
Diatomic	5	$\frac{5}{2}nRT$
Polyatomic	6	$3nRT$

7 UNIT CHECK

Quantity	Unit
Pressure p	Pa = N m^{-2}
Molar mass M	kg mol^{-1} (not g mol^{-1})
$\langle c^2\rangle$	m^2 s^{-2}
Internal energy U	J

8 COMMON SECTION A TRAPS

- Using °C instead of K.
- Using M in g mol^{-1} in $\sqrt{3RT/M}$.
- Confusing n (moles) with N (molecules).
- Mean KE depends only on T, not on the type of gas.
- Doubling T multiplies c_{rms} by $\sqrt{2}$, not 2.

EXAM TIPS

- ✓Convert cm^3 → m^3 (×10^{-6}) and kPa → Pa.
- ✓Write $n = m/M$ before using $pV = nRT$.
- ✓Use ratios: $c_{\text{rms}} \propto \sqrt{T/M}$.
- ✓List the assumptions in full sentences.